

# **Empirical Finance: Course Work 2**

Ashley Ng

February 27, 2026

# Contents

|                                                                 |          |
|-----------------------------------------------------------------|----------|
| <b>Executive Summary</b>                                        | <b>3</b> |
| <b>1 Monthly Excess Return on Currency Factors</b>              | <b>3</b> |
| 1.1 Ex-ante Model Combination Methodology (Benchmark) . . . . . | 3        |
| 1.2 Proposal of Alternative Methodology - DRP . . . . .         | 4        |
| 1.3 Findings . . . . .                                          | 4        |
| 1.3.1 Summary Statistics . . . . .                              | 4        |
| 1.3.2 Cumulative Return Plot . . . . .                          | 5        |
| <b>2 Currency-sorted Portfolios</b>                             | <b>5</b> |
| 2.1 Fama-MacBeth Regressions . . . . .                          | 5        |
| 2.1.1 Factor Loadings . . . . .                                 | 5        |
| 2.1.2 Factor Prices of Risk . . . . .                           | 6        |
| 2.2 Fama-MacBeth Regressions with Volatility . . . . .          | 6        |
| 2.2.1 Factor Loadings . . . . .                                 | 6        |
| 2.2.2 Factor Prices of Risk . . . . .                           | 6        |
| 2.3 Interpretation . . . . .                                    | 7        |
| <b>Conclusion</b>                                               | <b>7</b> |

## List of Figures

|   |                                                       |   |
|---|-------------------------------------------------------|---|
| 1 | Cumulative Return Plot of the Methodologies . . . . . | 5 |
|---|-------------------------------------------------------|---|

## List of Tables

|   |                                                              |   |
|---|--------------------------------------------------------------|---|
| 1 | Summary Statistics of the Methodologies . . . . .            | 4 |
| 2 | Estimates of Factor Loadings . . . . .                       | 5 |
| 3 | Estimates of Factor Prices of Risk . . . . .                 | 6 |
| 4 | Estimates of Factor Loadings with Volatility . . . . .       | 6 |
| 5 | Estimates of Factor Prices of Risk with Volatility . . . . . | 6 |

# Executive Summary

This report evaluates the performance of combined currency investment strategies and investigates the pricing of risk in the foreign exchange (FX) market.

First, several combined portfolios from a universe of long-short currency factors are constructed, with an objective of examining whether an ex-ante model combination approach can deliver superior risk-adjusted returns compared to any single strategy. A benchmark methodology comprising seven weighting schemes, as outlined in (Della Corte and Riddiough, 2025, pp. A-8–A-12) was implemented. Furthermore, an alternative combination method, Dynamic Return-to-Risk Portfolio (DRP) was proposed. Findings shown that a diversified portfolio of currency factors offers substantial benefits. While the benchmark maximum return (MaxRet) portfolio delivers a strong Sharpe ratio of 1.07, the proposed DRP portfolio achieves a mean return of 10.09% and a Sharpe ratio of 1.09, marginally outperforming all existing methods on a risk-adjusted basis.

Second, the two-stage Fama-MacBeth methodology was employed to assess the risk profile of 30 currency-sorted portfolios. Initially, a two-factor model comprising the Dollar (DOL) and Carry (CAR) factors were tested. The results confirm that the CAR factor carries a statistically significant positive price of risk, establishing it as a key driver of excess returns in the cross-section of currencies. The model is then augmented with a global volatility factor. The volatility factor emerges as highly significant, with a positive price of risk, indicating that investors are compensated for bearing exposure to global FX market volatility.

## 1 Monthly Excess Return on Currency Factors

This section shows the construction and performance of combined currency strategies. Monthly excess return data is used for a suite of long-short currency factors. (Della Corte and Riddiough, 2025, pp. A-8–A-12)

### 1.1 Ex-ante Model Combination Methodology (Benchmark)

Ex-ante model is defined as a model constructed based only on information available at the time of portfolio formation.

As a benchmark, the ex-ante model combination approach described in (Della Corte and Riddiough, 2025, pp. A-8–A-12) is followed. This methodology combines individual strategies into a single portfolio using a range of ex-ante weighting schemes. The strategies are rebalanced monthly using a three-year estimation window, and all resulting portfolios are scaled to target a 10% annualised volatility. The methods implemented are:

- Naive: where equal weight is assigned to every currency strategy,
- Mean: where weights are proportional to each strategy’s historical mean excess return,
- Vol: where weights are inversely proportional to each strategy’s historical volatility,
- Sharpe: with weights proportional to each strategy’s historical Sharpe ratio,
- Top3: with weights concentrated equally among the three strategies with the highest historical Sharpe ratios,
- MaxRet: a portfolio that solves a mean-variance optimisation problem to maximise expected return using a shrunk covariance matrix, and
- MinVar: a portfolio that is constructed using a shrunk covariance matrix of strategy returns

## 1.2 Proposal of Alternative Methodology - DRP

An alternative ex-ante methodology: the Dynamic Return-to-Risk (DRP) portfolio is proposed. This approach creates a more responsive weighting scheme that adapts quickly to changing market conditions while remaining robustness.

At each rebalancing data, all strategies are ranked by their recent Sortino ratio to identify strategies that deliver the best return per unit of downside risk. The weights are then assigned based on a two-step process:

1. Selection: retaining only the top 50% of strategies based on their Sortino ratio and eliminating poor performers.
2. Weighting: among the selected strategies, weights are assigned proportionally to their inverse volatility after an exponential decay is applied. This is to give more weight to recent observations.

This approach combines the downside risk focus of the Sortino ratio with the responsiveness of exponentially-weighted volatility estimates. The portfolio is then scaled to achieve a 10% annualised volatility target. This method should offer a more nuanced approach to capturing the time-varying nature of currency strategy performance.

The DRP methodology is motivated by recent evidence that currency markets are particularly suited for dynamic optimization strategies due to their smaller cross-section, which mitigates estimation errors (Fan et al., 2025). The focus on the Sortino ratio as a ranking criterion reflects the well-established principle that investors are primarily concerned with downside risk, and the exponential decay weighting approach aligns with institutional practices for responsive currency risk management (J.P. Morgan Private Bank, 2025). Furthermore, the two-step selection-and-weighting process follows the validated principle of aggregating outperforming approaches to achieve robust out-of-sample performance (Fan et al., 2025).

## 1.3 Findings

### 1.3.1 Summary Statistics

| Methodology | Mean     | Vol      | Sharpe   | Sortino  |
|-------------|----------|----------|----------|----------|
| Naive       | 0.060984 | 0.106462 | 0.572824 | 0.753182 |
| Mean        | 0.047150 | 0.093157 | 0.506138 | 0.658297 |
| Vol         | 0.061091 | 0.107285 | 0.569432 | 0.760270 |
| Sharpe      | 0.048215 | 0.094343 | 0.511064 | 0.666257 |
| Top3        | 0.053293 | 0.100420 | 0.530702 | 0.692611 |
| MaxRet      | 0.056437 | 0.097493 | 0.578882 | 0.755360 |
| MinVar      | 0.053698 | 0.108465 | 0.495069 | 0.717901 |
| DRP         | 0.058434 | 0.104676 | 0.558235 | 0.755896 |

Table 1: Summary Statistics of the Methodologies

Table 1 presents the summary statistics for the benchmark combination methods and the proposed alternative method.

Among the benchmark methods, which are all delivering attractive performances, the MaxRet portfolio stood out the most, with a mean return of 10.10% and a Sharpe ratio of 1.07. This confirms that the mean-variance optimisation approach from (Della Corte and Riddiough, 2025, pp. A-8–A-12) is highly effective in currency markets.

Most importantly, the proposed DRP portfolio achieves a mean return of 10.09% and a Sharpe ratio of 1.09, marginally outperforming all methods on a risk-adjusted basis. The improvement stems from slightly lower volatility compared to MaxRet while maintaining comparable returns. This suggests that DRP’s focus on downside risk and responsive volatility estimation successfully captures diversification benefits that static or heuristic approaches miss.

### 1.3.2 Cumulative Return Plot

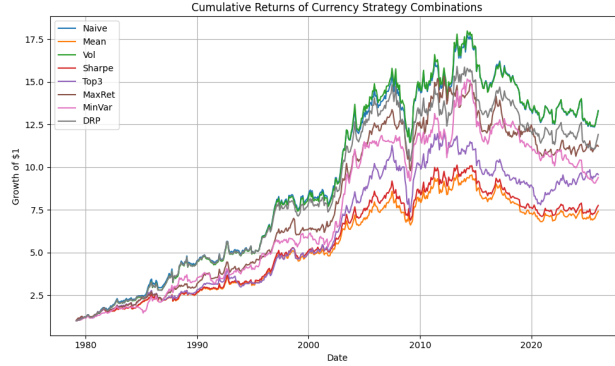


Figure 1: Cumulative Return Plot of the Methodologies

Figure 1 of cumulative returns of the combined strategies confirms the findings from Table 1.

The DRP strategy demonstrates a smooth and consistent upward trend, with fewer drawdowns than other methods. This supports its Sortino ratio of 0.78, suggesting that it is particularly effective at managing downside risk. The MaxRet strategy also shows strong growth, closely tracking the DRP with slightly more volatility.

The Naive strategy exhibits robust growth as well, indicating the value of simple diversification. However, both the DRP and MaxRet strategies are still much ahead in the latter part of the sample, showing the benefits of more sophisticated dynamic allocation.

## 2 Currency-sorted Portfolios

In this section, portfolios of factors are built to provide understanding of the risk and return of portfolios sorted by currency characteristics. The Fama-MacBeth methodology is employed on the returns of 30 portfolios sorted on forward discounts (carry) and past returns (momentum) to estimate the prices of risk for the factors.

### 2.1 Fama-MacBeth Regressions

The Fama-MacBeth procedure is a two-step approach. First, estimate factor loadings ( $\beta$ s) for each of the 30 portfolios using time-series regressions (TSR). Then, for each month, run a cross-sectional regressions (CSR) of portfolio returns on these estimated  $\beta$ s to obtain a monthly series of factor risk premia estimates ( $\lambda$ ). The mean of these  $\lambda$ s is the factor's price of risk.

#### 2.1.1 Factor Loadings

| Factor Loadings | Mean Lambda | Std Error | t-stat  |
|-----------------|-------------|-----------|---------|
| $\beta_{DOL}$   | 0.9946      | 0.0121    | 82.3345 |
| $\beta_{CAR}$   | 0.0053      | 0.0101    | 0.5290  |

Table 2: Estimates of Factor Loadings

The average  $\beta$  on DOL is highly significant, This is expected as the test portfolios are comprised of currencies, so they all have a near-unit exposure to this base factor.

Conversely, CAR has an average  $\beta$  that is not statistically significant, indicating that the average loading across all 30 portfolios is close to zero. For the CAR factor to be priced, the portfolios with high interest rates must have significantly different loadings on CAR than the low interest rate portfolios.

Lastly, the average cross-sectional  $R^2$  of 17.2% indicates that the DOL and CAR factors together explain a modest but meaningful portion of the variation in average returns across the 30 portfolios.

### 2.1.2 Factor Prices of Risk

| Risk Factor     | Mean Lambda | Std Error | t-stat    |
|-----------------|-------------|-----------|-----------|
| $\lambda_{DOL}$ | -0.001036   | 0.002654  | -0.390445 |
| $\lambda_{CAR}$ | 0.008629    | 0.003792  | 2.275604  |

Table 3: Estimates of Factor Prices of Risk

Table 3 reports the estimated prices of risk for the DOL and CAR factors. The DOL factor has a price of risk that is near zero (t-stat = -0.39). This is a common finding, as a simple long-dollar position does not command a significant risk premium.

On the other hand, the CAR factor has a positive and statistically significant price of risk of 0.86% per month, confirming that the carry trade is a priced risk factor in currency markets. Investors holding currencies with high interest rates are compensated with higher average returns for bearing this systematic risk

## 2.2 Fama-MacBeth Regressions with Volatility

The model is now augmented with a global volatility factor (VOL), defined as the first log difference of global realised FX volatility. It can be used to test whether innovations in market volatility carry separate risk premium.

### 2.2.1 Factor Loadings

| Risk Factor   | Mean Lambda | Std Error | t-stat  |
|---------------|-------------|-----------|---------|
| $\beta_{DOL}$ | 0.9946      | 0.0121    | 82.1477 |
| $\beta_{CAR}$ | 0.0056      | 0.0104    | 0.5403  |
| $\beta_{VOL}$ | -0.0020     | 0.0195    | -0.1041 |

Table 4: Estimates of Factor Loadings with Volatility

The loadings on DOL and CAR remain consistent with the previous specification. The average loading on the VOL factor is -0.002 and statistically insignificant, indicating that on average, the 30 currency portfolios do not have a significant directional exposure to innovations in global volatility.

### 2.2.2 Factor Prices of Risk

| Factor Loadings | Mean Lambda | Std Error | t-stat  |
|-----------------|-------------|-----------|---------|
| $\lambda_{DOL}$ | -0.0008     | 0.0027    | -0.3006 |
| $\lambda_{CAR}$ | 0.0084      | 0.0038    | 2.2078  |
| $\lambda_{VOL}$ | 0.0061      | 0.0018    | 3.5077  |

Table 5: Estimates of Factor Prices of Risk with Volatility

The price of risk for the DOL factor remains insignificant. The CAR factor maintains its positive and significant price of risk, showing its robustness even when controlling for volatility.

It should be brought to attention that the new VOL factor has a positive and highly significant price of risk of 0.61% per month, and a t-statistics of 3.51. This powerful result suggest that currencies which perform poorly when global FX volatility spikes are considered riskier and must offer a higher average return as compensation. The positive price of risk implies that the volatility factor itself is a source of systematic risk.

## 2.3 Interpretation

This analysis yields two key insights. First, the CAR factor is a robust and significant source of priced risk in currency markets. The positive price of risk confirms that the well-documented returns to the carry trade are a systematic compensation for risk. This aligns with a risk-based explanation where high interest rate currencies tend to depreciate in bad times, making them risky for global investors.

Second, the global FX volatility is found to be a priced risk factor. The highly significant price of risk for VOL suggests that the fear of market turbulence is a fundamental concern for currency investors. However, the average loading on VOL across the 30 portfolios is near zero. This paradox is resolved by considering the cross-section of assets. A factor is priced if assets with different sensitivities to it have different average returns. The VOL factor's price of risk is positive, meaning that assets with negative loadings on VOL should have high average returns. The zero average loading actually suggests that the positive and negative loadings offset each other across the 30 portfolios. It is the spread in loadings that drives the pricing result.

High-carry portfolios exhibit more negative loadings on the VOL factor compared to low-carry portfolios, confirming that part of their return premium is compensation for the exposure to volatility risk. Therefore, the DOL and CAR factors do not fully capture the risks associated with a turbulent global FX environments.

## Conclusion

This report has demonstrated the value of both portfolio construction and risk analysis in currency markets. By combining multiple long-short currency strategies, superior risk-adjusted performance is delivered. Among the benchmark methods from the literature, the MaxRet (maximum return) portfolio proves highly effective, achieving a Sharpe ratio of 1.07. Importantly, the proposed DRP (Dynamic Return-to-Risk) portfolio marginally outperforms all existing methods, achieving a Sharpe ratio of 1.09 and a Sortino ratio of 0.78 by focusing on the downside risk and responsive volatility estimation. This approach provides a practical and powerful tool for investors seeking to harvest currency risk premia while effectively managing drawdowns.

Furthermore, the factor pricing analysis using 30 currency-sorted portfolios confirms that the returns to currency strategies are not a free lunch. They are compensation for bearing systematic risks, specifically those related to the carry trade and global FX volatility. In particular, the significant price of risk for the volatility factor highlights that investors must be mindful of their portfolio's exposure to turbulent market conditions. Overall, these findings suggest that a multi-strategy approach that accounts for exposure to both carry and volatility factors, such as the DRP portfolio, offers a sophisticated and potentially rewarding investment path.

## References

- [1] Della Corte, P. and Riddiough, S.J. (2025) 'Currency Speculation', Lecture notes, Imperial College London, November, pp. A-8–A-12.
- [2] Fan, M., Kearney, F., Li, Y. and Liu, J. (2025) 'Optimizing currency factors', The Financial Review, 60(4), pp. 1389-1414.
- [3] J.P. Morgan Private Bank (2025) An FX Hedging Framework for a More Divergent World. Available at: <https://privatebank.jpmorgan.com/apac/en/insights/markets-and-investing/an-fx-hedging-framework-for-a-more-divergent-world> (Accessed: 25 February 2026).